

School of Computer Science Engineering and Technology

Course- B.Tech.	Type- Core
Course Code- 2025CSET106	Course Name- Discrete Mathematical Structures
Year- I	Semester- Even (2 nd Semester)

Tutorial-4

Tutorial No.	Name	CO1	CO 2	CO3
4	Methods of Proofs	✓	--	✓

Objective: The main objective of this tutorial sheet is to learn the methods of proof and mathematical induction.

Questions:

1. By using Principle of Mathematical Induction, show that n^3+2n is divisible by 3 for all positive integers.
2. By using Principle of Mathematical Induction, show that 8^n-3^n is divisible by 5 for all positive integers.
3. By using Principle of Mathematical Induction, for all positive integers show that

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

4. Show that $2^n > n^3$ for $n \geq 10$.
5. Using method of direct proof show that for all positive integers if n is **even** then n^2 is **even**.
6. Using method of direct proof show that difference of **square of two consecutive integers** is an **odd integer**.
7. Using indirect proof show that $3n+2$ is odd, then n is odd.